

Mohammed Mahdi Jahangiri

AI Engineer / Machine Learning Engineer

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AI Engineer with published research on LLM-driven dialogue systems (ICHORA 2024) and hands-on experience building NLP pipelines, fine-tuning models, and deploying AI via REST APIs. 2+ years of production software engineering in Agile teams with code review and cross-functional coordination. MSc dissertation focuses on safe AI companions using RAG and local LLM inference.

Technical Skills

Core Languages: Python, C++, C#
ML & AI: PyTorch, Hugging Face Transformers, scikit-learn, pandas, numpy, NLP, LLM fine-tuning, RAG, model evaluation, behaviour trees
APIs & Data: FastAPI, REST APIs, SQL, JSON, Wit.ai, Whisper
Engineering: Git, code review, Agile/Scrum, profiling and optimisation, cross-functional coordination

Selected Projects & Publications

AI-Driven NPC Interaction Prototype Unreal Engine 5, C++, Python, FastAPI

- Built a local **Python FastAPI** server bridging real-time microphone input to a **C++** application via a custom REST API
- Integrated **Whisper**, **Wit.ai** and local **Hugging Face** models for real-time text and voice emotion detection
- Trained the voice-emotion model and configured **Wit.ai** intents using custom voice recordings
- Wired microphone capture in **C++**, returning predicted intent/emotion signals to drive real-time **behaviour-tree** decision logic

MSc Thesis — Safe AI Companion Systems Python, JSON, LLM Fine-Tuning, RAG

- Built a custom AI companion with **RAG**-style context retrieval, using **JSON** request/response files and a **Python** LLM bridge
- Built a local LLM pipeline with bounded context, safety constraints and prompt-engineering guardrails
- Evaluated AI agent against branching-dialogue baselines, measuring interaction quality, latency, safety and reliability

Publication — ICHORA 2024 Balancing Game Satisfaction and Resource Efficiency: LLM and Pursuit Learning Automata for NPC Dialogues

- Published and presented paper proposing a resource-efficient local LLM pipeline for dynamic NPC dialogue
- Applied pursuit learning automata to balance player satisfaction against computational cost

Experience

Engineering Team Coordinator — Landell Games Feb 2025 – Present Remote / Sweden Promoted from Engineering Intern (Dec 2024 – Jan 2025). Team lead on a live-service software project.

- Coordinated a **3–6 person engineering team**, translating production priorities into sprint tasks and raising sprint success from **40% to 90%**
- Profiled and optimised system performance bottlenecks, improving frame time by **10–30 FPS** on target hardware
- Conducted **code reviews** and enforced clean **Git/Plastic** branch hygiene across a remote team
- Owned weekly build pipelines and mediated cross-functional concerns between **5 department leads**

Education

MSc Artificial Intelligence — Cardiff University 2025–2027 Taught component completed June 2026. Dissertation in progress: safe AI companions using RAG and local LLM inference.

- Computer Vision: bounding-box localisation via **RF-DETR** fine-tuning on **VOC2012** using **PyTorch**
- Computational Linguistics: **NLP** pipeline development with data gathering, labelling and parameter tuning using **pandas** and **scikit-learn**; speech-to-text, intent detection and emotion analysis
- Technical AI: real-time voice-driven AI pipeline with **FastAPI**, **Whisper** and **Hugging Face**, integrated into a **C++** application environment
- AI Safety Founding Fellow (Spring 2026):** Completed fundamentals fellowship; evaluated catastrophic AI risks, model evaluations, privacy, and AI governance

BEng Computer Engineering — Islamic Azad University, Tehran 2020–2025 Graduated with **First-Class Honours**; core study: **Data Structures & Algorithms**, **Advanced Programming (C++)**, **SQL** and database design, Computational Architecture and Networks

Awards & Interests

DiffTrailblazers — Cardiff University: Won 1st place, receiving a **£1,500** prize for a practical innovation project.
Interests: Judo (green belt) | Mandarin Chinese | XR R&D